

2024 Semester One (June 2024) Examination Period

Faculty of Science

EXAM CODES: MAT9004
TITLE OF PAPER: Mathematical foundations for data science and AI
EXAM DURATION: 3 hours 10 mins

Rules

By starting your eExam, you're confirming that you're well enough to do the exam. If you become unwell during the assessment you must alert your online supervisor or contact the Exam Support hotline.

During your eExam, you must not have in your possession any item/material that has not been authorised for your exam. This includes books, notes, paper, electronic device/s, smart watch/device, or writing on any part of your body. Authorised items are listed above. Items/materials on your device, desk, chair, in your clothing or otherwise on your person will be deemed to be in your possession. Mobile phones must be switched off and placed face-down on your desk during your exam attempt.

You must not retain, copy, memorise or note down any exam content for personal use or to share with any other person by any means during or following your exam. You are not allowed to copy/paste text to or from external sources unless this has been authorised by your Chief Examiner.

You must comply with any instructions given to you by Monash exam staff.

As a student, and under Monash University's Student Academic Integrity procedure, you must undertake all your assessments with honesty and integrity. You must not allow anyone else to do work for you and you must not do any work for others. You must not contact, or attempt to contact, another person in an attempt to gain unfair advantage during your assessment. Assessors may take reasonable steps to check that your work displays the expected standards of academic integrity.

Failure to comply with the above instructions, or attempting to cheat or cheating in an assessment may constitute a breach of instructions under regulation 23 of the Monash University (Academic Board) Regulations or may constitute an act of academic misconduct under Part 7 of the Monash University (Council) Regulations.

Authorised Materials

CALCULATORS	☐ YES	☒ NO
DICTIONARIES	☐ YES	☒ NO
NOTES	☐ YES	☒ NO
WORKING SHEETS	☒ YES	☐ NO
PERMITTED ITEM	☐ YES	☒ NO

if yes, items permitted are:

Instructions

Once your exam finishes, you will be given time to scan a QR code and upload your answers using  your smartphone and laptop.

[Here's how to do it.](#) 

- There are two types of questions: "short answer questions" and "hand-written response".
- Unless otherwise specified, each answer to a "short answer question" on this exam is a rational number; enter your answers as follows. If the answer is an integer then enter it as a numeral. For example 16, 1, or -1 but NOT sixteen, 1.0, minus 1. If the answer is a non-integer rational number then enter it as [numerator]/[denominator] where the fraction is in lowest terms, [numerator] and [denominator] are integers entered as above. For example, -1/2, 5/4 or 11/16 NOT 1/-2, 1.25 or 22/32. Importantly, no answer to a "short answer question" should contain a space.
- The hand-written responses will require you to upload your written work using a camera (via a phone or tablet) in the 30 minutes after the exam is over. The procedure for doing this will be explained to you at the end of the exam.
- Note: Please widen your browser window to see the questions displayed in full size.
- A formula sheet is provided within the exam, before the questions (on page 2).
- You are not allowed to consult books or online sources during the exam. You may use an unlimited number of blank A4 paper as working sheets.
- Once the exam duration is finished, your exam will automatically submit. Please ensure you finalise your answers before the end of the allocated exam time.

When your exam duration has ended, you are not permitted to write or edit any of your answer sheets. Please note you must remain under exam conditions and are not permitted to adjust your answer sheets during the upload process once you have submitted your exam.

In this assessment, you must **not** use generative artificial intelligence (AI) to generate any materials or content in relation to the assessment task.

Instructions

Information

You can review your exam instructions by clicking the 'Show Instructions' button above.

Information

Getting help and support

Assessments without online supervision

If you have any difficulties (technical, medical or otherwise), call the Assessment Support hotline.

If your internet connection drops out during your assessment, don't panic – your responses are automatically saved every 30 seconds.



Assessment Support

Australia: +61 3 9905 4300
Malaysia: +60 3 5514 5600
China: +800 666 274 73





Assessments with online supervision

If you have any difficulties (technical, medical or otherwise) use the Raise Hand button or webchat function to alert your online supervisor.

If you've lost your connection to your online supervisor, call the Assessment Support hotline.

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Formula Sheet

Information

$$a^0 = 1, \quad a^x a^y = a^{x+y}, \quad a^{xy} = (a^x)^y.$$

$$\log_a(1) = 0, \quad \log_a(xy) = \log_a(x) + \log_a(y), \quad \log_a(x^y) = y \log_a(x), \quad \log_a(x) = \frac{\log_b(x)}{\log_b(a)}.$$

$$\text{If } f(x) = x^n, \text{ then } f'(x) = nx^{n-1}.$$

$$\text{If } f(x) = a^{cx+d}, \text{ then } f'(x) = c \ln(a) a^{cx+d}.$$

$$\text{If } f(x) = \log_a(cx + d), \text{ then } f'(x) = \frac{c}{\ln(a)(cx+d)}.$$

$$\nabla f(x, y) = \begin{bmatrix} f_x(x, y) \\ f_y(x, y) \end{bmatrix}$$

$$H(x, y) = \begin{bmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{bmatrix}.$$

$$\det(H(x, y)) > 0 \text{ and } f_{xx}(x, y) > 0 \implies \text{local minimum.}$$

$$\det(H(x, y)) > 0 \text{ and } f_{xx}(x, y) < 0 \implies \text{local maximum.}$$

$$\det(H(x, y)) < 0 \implies \text{saddle point.}$$

$$\det(H(x, y)) = 0 \implies \text{test inconclusive.}$$

$$\text{If } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ then } \det(A) = ad - bc \text{ and } A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Selections of r elements
from a set of n elements:

	Unordered	Ordered
With repetition	$\binom{n+r-1}{r}$	n^r
Without repetition	$\frac{n!}{r!(n-r)!} = \binom{n}{r}$	$\frac{n!}{(n-r)!}$

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

$$\Pr(A|B) = \frac{\Pr(B|A)\Pr(A)}{\Pr(B|A)\Pr(A) + \Pr(B|\bar{A})\Pr(\bar{A})}.$$

$$\text{Discrete uniform: } \Pr(X = k) = \frac{1}{b-a+1} \text{ has } E[X] = \frac{a+b}{2} \text{ and } \text{Var}[X] = \frac{(b-a+1)^2-1}{12}.$$

$$\text{Bernoulli: } \Pr(X = k) = \begin{cases} p & \text{for } k = 1 \\ 1-p & \text{for } k = 0 \end{cases} \text{ has } E[X] = p \text{ and } \text{Var}[X] = p(1-p).$$

$$\text{Geometric: } \Pr(X = k) = p(1-p)^k \text{ has } E[X] = \frac{1-p}{p} \text{ and } \text{Var}[X] = \frac{1-p}{p^2}.$$

$$\text{Binomial: } \Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \text{ has } E[X] = np \text{ and } \text{Var}[X] = np(1-p).$$

$$\text{Poisson: } \Pr(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \text{ has } E[X] = \lambda \text{ and } \text{Var}[X] = \lambda.$$

$$\text{Continuous uniform: } \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise.} \end{cases} \text{ has } E[X] = \frac{a+b}{2} \text{ and } \text{Var}[X] = \frac{(b-a)^2}{12}.$$

$$\text{Exponential: } \lambda e^{-\lambda x} \text{ has } E[X] = \frac{1}{\lambda} \text{ and } \text{Var}[X] = \frac{1}{\lambda^2}.$$

$$\text{Normal: } \frac{1}{\sqrt{2\sigma^2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \text{ has } E[X] = \mu \text{ and } \text{Var}[X] = \sigma^2.$$

$$t\text{-statistic: } t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Foundations

Question 1

Evaluate $\sum_{j=2}^5 (2j - 1)$

1.5
Marks

Question 2

Find the value of the following.

$$\prod_{j=2}^3 (j^2 + 2)$$

1.5
Marks

Question 3

Choose which of the following terms correctly applies to the function $f(x) = 3x^3 - 1$

2
Marks

Select one:

- ☐ a. Convex
- ☐ b. Concave
- ☐ c. Both convex and concave
- ☐ d. Neither convex nor concave

Question 4

Find the equation of the line through the points $(6, -2)$ and $(4, 8)$. The equation of the line should be of the form $y = mx + b$.

m =

b =

2
Marks

Question 5

Suppose $f(x) = bx^a$, and that a log-log plot (with logarithms taken base 2) of this function shows a straight line with slope -4 and y-intercept 4. What are a and b ?

a =

b =

2
Marks

Functions

Question 6

Consider the function $f : [0, \infty) \rightarrow \mathbb{R}$ where $f(x) = x^2 + 2x$. What is the value of $f^{-1}(24)$?

2
Marks

Question 7

If $f(x) = 2(e^{2x} + 3)^{1/2}$, find $f'(0)$.

2
Marks

Question 8

If $f(x) = (-x^2 - 5)e^{-x}$, find the slope of the tangent line to the graph of f at $x = 0$.

2
Marks

Question 9

Find the maximum value of the product xy , under the constraint that $x + 3y = 60$.

3
Marks

Question 10

For the function $h(x) = \ln(x^2 - 6x + 10)$, find x at which h has a stationary value.

2
Marks

Question 11

Evaluate $\int_{-1}^3 (-3x^2 + 4x + 2)dx$

2
Marks

Question 12

Find the stationary point of the function $f(x, y) = xy - y^2 + 2x$.

Make sure to express the point in the form (a, b) .

3
Marks

Question 13

For the function $g(x, y) = \ln(x^2 + y) + xy^2$, find the gradient vector ∇g at the point $(1, -2)$.

Make sure to express your vector in the form (a, b) .

3
Marks

Question 14

If the gradient of a function f satisfies

$$\nabla f(x, y) = \begin{pmatrix} 4x^3y^2 - 4xy + 2xy^2 \\ 2x^4y - 2x^2 + 2x^2y \end{pmatrix},$$

find the determinant of $H(-1, 1)$, where H refers to the Hessian matrix.

2
Marks

Question 15

What is the residual sum of squares (RSS) when approximating $(-1, 5), (2, 1), (6, 3)$ by $f(x) = -x + 5$?

3

Marks

Linear algebra

Question 16

Let $A = \begin{pmatrix} 4 & 6 \\ x & 12 \end{pmatrix}$. For which $x \in \mathbb{R}$ is A not invertible?

2
Marks

Question 17

Let $A = \begin{pmatrix} -3 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 2 & -1 \end{pmatrix}$. Then $A^{-1} = \begin{pmatrix} 1/2 & 2 & -3/2 \\ 1/2 & 1 & -1/2 \\ 3/2 & x & -7/2 \end{pmatrix}$. Find x .

2
Marks

Question 18

Let $A = \begin{pmatrix} -2 & -4 & 2 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{pmatrix}$. One of the eigenvalues of A is 3; if the corresponding eigenvector is of the form $(x, y, 1)$, find x and y .

2
Marks

x=

y=

Question 19

Let $a = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$ and $b = \begin{pmatrix} 3 \\ x \\ 4 \end{pmatrix}$. Suppose that a and b are orthogonal. What is x ?

2
Marks

Question 20

The length of the vector $(-8, 9, -12)$ is an integer. Find that integer.

2
Marks

Question 21

Solve the following system of linear equations using Gaussian elimination:

$$\begin{pmatrix} 1 & -1 & 1 \\ -1 & -1 & 1 \\ 3 & -2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ -2 \\ 9 \end{pmatrix}$$

x=

y=

z=

3
Marks

Combinatorics

Question 22

Members of a hiring panel are asked to rank who they believe are the top three applicants for a job, specifying who is first, second, and third. There are 12 applicants. How many different ways are there to rank them in this manner?

2

Marks

Question 23

A volleyball team must choose 6 players to be on the court at any given time. If there are 11 players on the team, how many different ways are there to choose a group of 6 players?

2

Marks

Question 24

There are 27 letters in the Spanish alphabet. Suppose we form a password 7 letters long using only lower case letters from the Spanish alphabet. There are exactly 3^n such passwords. Find n .

2

Marks

Question 25

At a pizza shop there are 5 different types of pizza. How many different purchases of 6 pizzas can one make?

2

Marks

Probability and graphs

Question 26

X is a discrete random variable defined to be the value when a positive integer is chosen randomly and uniformly from 1 to 5, and Y is a discrete random variable defined to be the value when a positive integer is chosen randomly and uniformly from 1 to 7. Assume that X and Y are independent. Find the variance $\text{Var}(2X + 3Y)$.

2
Marks

Question 27

A game is played in which a fair coin is tossed three times. If an even number of heads are thrown the player wins 12 dollars, if one head occurs the player wins 16 dollars and if three heads occur the player loses 8 dollars. What is the expected value of their winnings, in dollars? Note: just put the number, do not put a dollar sign. Also, do not forget that 0 is an even number!

3
Marks

Question 28

Suppose a discrete random variable X has a probability function $h(x)$ defined by the following table.

3
Marks

x	3	7	11
Pr(X=x)	1/4	1/4	1/2

Find $E[X]$.

Question 29

A graph has a degree sequence of $(1, 1, 2, 3, 3, 3, 3)$. How many edges does it have?

2
Marks

Question 30

3
Marks

The graph G has the adjacency matrix

$$M = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

How many vertices does G have?

How many edges does G have?

How many components (i.e. separate pieces) does G have?

Question 31

3
Marks

Suppose a random variable X admits the pdf $h(x) = cx^2$ for $|x| < 1$ and 0 elsewhere. First determine the constant c , and then use this to find $E[X]$ and $\text{Var}(X)$; remember to reduce all fractions, and leave numbers as improper fractions rather than mixed numbers or decimals.

$c =$

$E[X] =$

$\text{Var}(X) =$

Long answer questions

Question 32

Let $f(x) = x^2 e^{-x^2}$. Find all stationary points of f on its domain. Classify the stationary points as either local maximums or local minimums. Explain your method clearly.

6
Marks



Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have **extra time** to access your phone to scan a **QR code** and **upload** your answer.
- Clearly label each page with **Student ID** and **this question number** (and **sub part** if applicable) (for example, 'Question 7a')
- **Do not** write your Name on it

No. of answer sheets: 3

Question 33

Let $f(x, y) = 2x^2y - xy^2 + 6x^2 + 9x$. Determine the gradient, i.e. ∇f , and hence find the Hessian matrix of $f(x, y)$. Find the stationary points, and use the Hessian to classify as much as possible the nature of the stationary points of f (note: if the Hessian test fails just say so, you do not need to try other tests).

6
Marks



Please answer the question on your blank piece of paper.

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- **Do not** write your Name on it

No. of answer sheets: 3

Question 34

Consider the following matrix.

$$A = \begin{bmatrix} 3 & -2 \\ 6 & -5 \end{bmatrix}$$

- Find the eigenvalues of A .
- Find eigenvectors corresponding to each of the eigenvalues.
- Find matrices P , D with P invertible and D diagonal so that $A = PDP^{-1}$. You do not need to calculate P^{-1} .

6
Marks



Please answer the question on your blank piece of paper.

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- Clearly label each page with **Student ID** and **this question number** (and **sub part** if applicable) (for example, 'Question 7a')
- **Do not** write your Name on it

No. of answer sheets: 3

Question 35

6
Marks

Consider the following system of linear equations:

$$x - y + 2z + w = 4$$

$$2x - 2y + 3z + 4w = 10$$

$$x - y + 3z - w = 2,$$

Write this system of equations in a matrix form. Solve the system and write down all solutions (x, y, z, w) .



Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have **extra time** to access your phone to scan a **QR code** and **upload** your answer.
- Clearly label each page with **Student ID** and **this question number** (and **sub part** if applicable) (for example, 'Question 7a')
- **Do not** write your Name on it

No. of answer sheets: 3

Question 36

6
Marks

A life insurance company issues standard and preferred policies. Of the company's policyholders of a certain age, 60% have standard policies and a probability 0.01 of dying in the next year, while 40% have preferred policies and a probability of 0.008 of dying in the next year.

- a) Suppose a policyholder is chosen at random. What is the probability that they will die over the coming year?
- b) One of their policyholders who died in the most recent year is selected uniformly at random. What is the probability that the deceased held a standard policy?



Please answer the question on your blank piece of paper.

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- **Do not** write your Name on it

No. of answer sheets: 3